

## Scenario Context for Spring 2026

### DhakaPark Multi-Site Parking Management

DhakaPark Ltd. is a growing parking space rental business operating across Dhaka city. The company manages **12 parking locations** in high-demand zones such as Gulshan, Banani, Dhanmondi, Mirpur, Motijheel, Uttara, Farmgate, and Mohammadpur. Some sites are **open-air lots**, some are **basement/covered**, and two are **multi-level structures**. DhakaPark's customers include (1) **walk-in drivers** looking for short-term parking, (2) **monthly subscribers** (office workers and residents), and (3) **corporate clients** who rent a block of spaces for staff and visitors. DhakaPark also supports **event-based parking** (weekend fairs, concerts near venues) where pricing changes by time and demand. Each site has a small team: a site supervisor, cash collectors, gate attendants, and security guards. At head office (located in Tejgaon), there are teams for operations, finance, customer support, and management.

Over time, DhakaPark expanded quickly by leasing land from multiple owners and partnering with commercial buildings. However, the operational systems did not evolve. Each site currently works with a **semi-manual process**. At entry, attendants record vehicle details in a paper logbook (sometimes a local Excel file), issue a handwritten token, and collect cash in many cases. At exit, they calculate parking fees manually based on rough time estimates (entry time is not always recorded accurately), and they often face disputes with customers. Some sites use basic barcode cards for monthly subscribers, but the scanner devices are inconsistent and not linked to a central system. Customer complaints have increased: drivers frequently report **long queues, incorrect charges, lost tokens, and poor communication about availability** (customers reach a full lot and have to turn back). Corporate clients complain about lack of transparent usage reports and difficulty managing guest parking.

DhakaPark's leadership also faces severe visibility problems. Management cannot reliably answer basic questions such as: *Which sites are full at what times? Which sites generate the highest revenue? How many vehicles entered/exited yesterday across all locations? What is the average parking duration by location? How many monthly subscribers are active, and how many renew late?* Each site submits a daily cash summary to HQ via WhatsApp photos or handwritten notes. Finance staff manually re-enter these numbers into spreadsheets. Errors are common, and sometimes suspicious: cash shortfalls, mismatched counts, and inconsistent "free exit" adjustments. When a site supervisor is absent or replaced, recordkeeping becomes even worse. DhakaPark suspects revenue leakage due to manual collection and the inability to audit transactions end-to-end.

The company also struggles with **space utilization and planning**. Some sites are crowded during office hours but empty at night; others are busy on weekends. The business wants to introduce

**dynamic pricing** and subscription packages to improve occupancy and profitability. However, they cannot do this confidently because the data is incomplete and unreliable. The operations team also needs better control over staffing, shift scheduling, and incident reporting (e.g., minor vehicle damage, disputes, security issues). Currently, incidents are logged informally, often without photos, without timestamps, and without structured categorization. This becomes a legal and customer service problem when disputes happen later.

DhakaPark now plans to modernize with a single “DhakaPark Integrated Parking Information System” that supports both **customer-facing** and **internal operational management** needs. The goal is to reduce entry/exit time, improve accuracy and trust, strengthen revenue control, improve customer experience, and give management real-time visibility across sites. The new system must support:

- **Customer-facing services:** finding parking availability by location, pricing information, subscription signup/renewal, digital receipts, complaints, and support.
- **Operational services:** vehicle entry/exit capture, automated fee calculation, payment capture (cash + digital), capacity tracking, staff roles, audit logs, and incident handling.
- **Management services:** performance reporting, revenue summaries, occupancy analysis, peak-time insights, subscription analytics, customer satisfaction trends, and site-to-site comparison dashboards.

Because DhakaPark operates multiple sites with different constraints, the system must consider connectivity and device realities. Some locations have unstable internet, some use basic Android phones, and not all gates can install expensive hardware. DhakaPark is open to using **state-of-the-art transaction capture technologies** where feasible.

DhakaPark wants the modernization project delivered in **one defined phase** for this semester’s scope, but the phase must still feel realistic and valuable. The company expects your student teams (acting as vendors responding to an RFP) to propose the full solution design for that phase, including the TPS design, example MIS reports, an ESS dashboard concept, and the supporting technology/data architecture. DhakaPark also wants teams to propose improvements in **complementary assets** that will determine success.

Finally, DhakaPark wants to explore **AI/ML opportunities** that can add value once reliable data is available. Teams must propose AI/ML ideas that are realistic and clearly connected to the core system.